

AAA Frameworks Analysis

	Note(s)	Issue(s)	Recommendation(s)
Authorization /authentication	• Forensic Evidence: <i>The event log (ID 1227) shows a payroll event added to "FAUX_BANK" on 10/03/2023 at 8:29:57 AM.</i> • Threat Actor Identity: <i>The user is identified as "Legal/Administrator" accessing the system from computer "Up2-NoGud" with IP address 152.207.255.255</i>	• Stale Account Exploitation: <i>The account belongs to Robert Taylor Jr., a Legal Attorney whose contract ended on 12/27/2019. The account remained active for nearly four years after his departure.</i> • Authorization Failure: <i>The user maintained "Admin" status, which granted him the permissions to modify payroll data without a</i>	• Technical Control: <i>Implement Multi-Factor Authentication (MFA) for all administrative and financial accounts to prevent unauthorized access via compromised credentials.</i> • Operational Control: <i>Establish a strict Account De-provisioning Policy that requires the IT department to disable contractor access within 24 hours of their contract end date.</i>

		<i>second level of approval.</i>	• Managerial Control: Conduct Quarterly Access Reviews to identify and remove "orphan" or "stale" accounts that no longer have a business justification.
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Summary of Skills Demonstrated:

- Log Intelligence & Accounting: Performed forensic analysis of system event logs to reconstruct an unauthorized financial event.
- Identity Access Management (IAM): Identified critical vulnerabilities related to the "Principle of Least Privilege" and the mismanagement of contractor account lifecycles.
- Strategic Risk Mitigation: Developed actionable technical and operational recommendations, including MFA and automated de-provisioning policies, to harden the organization's security posture.

Access Control Investigation & AAA Mitigation Report Objective & Scenario

As the lead cybersecurity professional for a growing business, I was tasked with investigating a suspicious financial transaction (Event ID: 1227) involving an unauthorized deposit to an unknown account labeled "FAUX_BANK". The goal of this investigation was to perform a detailed "accounting" of the incident, identify the threat actor, uncover the exploited access control vulnerabilities, and recommend technical and operational mitigations.

Appendix: Forensic Evidence

Evidence A: System Event Log

The following log entry confirms the unauthorized addition of the FAUX_BANK payroll event:

- Event ID: 1227
- User: Legal/Administrator
- Computer: Up2-NoGud
- IP: 152.207.255.255
- Description: Payroll event added. FAUX_BANK

Evidence B: Employee Directory Reference

The directory confirms that the administrator account used in the breach was assigned to a contractor who left the organization nearly four years prior:

- Name: Robert Taylor Jr.
- Role: Legal Attorney (Contractor)
- End Date: 12/27/2019
- Authorization: Admin

Conclusion:

This investigation successfully identified the root cause of the unauthorized "FAUX_BANK" deposit as a failure in the organization's account lifecycle management. By exploiting a stale administrative account, the threat actor bypassed existing authorization controls. Implementing the recommended AAA enhancements, specifically Multi-Factor Authentication and automated de-provisioning, will significantly reduce the risk of future unauthorized access and ensure that organizational resources remain accessible only to active, authorized personnel.